Machine Learning for Customer Segmentation Using K-Means Clustering

Question: How can K-Means clustering help businesses group customers based on behavior and spending?

1. Introduction: Why Customer Segmentation Matters

Customer segmentation helps a business get a clearer picture of who it's trying to reach. By grouping people based on things like their demographics, interests, or buying behavior, a company can tailor its marketing so each segment sees messages that feel relevant to them.

Businesses can collect a large amount of information about their customers, including what they buy and how often they shop. Since customers can have different needs and habits, it can be difficult for a business to examine everyone individually. When the amount of customer data becomes large, it becomes too hard to sort through by hand. Finding patterns in this information can help businesses understand their customers and make better decisions.

Machine learning can help businesses understand their customers better. One machine learning technique that can be used for this is clustering, which groups people based on similarities. Instead of customer groups, clustering can find them automatically. These groups can help businesses improve their marketing and keep customers interested.

K-Means is an unsupervised machine learning algorithm that groups unlabeled data into K distinct, non-overlapping clusters. K-means is a way for a computer to sort things into K different groups. The "K" in K-Means represents the number of clusters (or groups) that the algorithm will partition the dataset into. For customer segmentation, K-Means can be used to automatically group customers into clear, separate groups based on things they have in common and how they act, without needing any labels beforehand.

This could help businesses send better ads, make more money, and keep customers coming back. This article will examine what K-Means is, how it groups people, and why it's useful.

2. Machine Learning, Unsupervised Learning, and Clustering

Machine learning is a branch of artificial intelligence that uses algorithms to learn patterns from example data. Once it learns these patterns, it can make accurate predictions about new, unseen data. IBM explains that machine learning helps computers recognize patterns on their own. After training on example data, the computer can spot similar patterns in new information and make decisions or predictions without being told exactly what to do.

Unsupervised learning is a type of machine learning that works with unlabeled data. This means the data has no predefined categories. The algorithm looks for hidden patterns and naturally forms groups, or clusters, without human guidance.

A simple example is the photo gallery on your phone. You have many pictures, but you never label them as "friends," "vacations," or "pets." With unsupervised learning, your

phone can automatically group similar photos together. It might collect all your dog pictures, beach photos, or selfies, even though you never organized them yourself.

Clustering is one method used in unsupervised learning. It groups data points that share similar traits. For customer information, clustering can group people who behave alike, such as customers who buy similar products or shop at similar times. Once these groups are formed, businesses can study them to understand customer behavior and make smarter decisions about marketing and products.

3. What Is K-Means Clustering?

K-Means is an unsupervised clustering algorithm, which means it groups data without using labels. Its job is to split the data into a set number of clusters, and you choose that number ahead of time. The K in K-Means stands for how many clusters you want. So if you pick K = 3, the algorithm will try to create three groups based on similarities in the data.

Each group has a center called a centroid. A centroid is basically the average position of all the data points in that cluster. As the algorithm works, these centroids move around to better represent the points in their group.

K-Means is especially helpful for numerical customer data, like how much customers spend or how often they shop. It can uncover natural patterns in this information and show different types of customer groups, such as high spenders or frequent shoppers. Businesses can use these groups to understand customer behavior and make smarter decisions about marketing and products.

4. How K-Means Works

K-Means clustering uses a step-by-step process to form groups in a dataset.

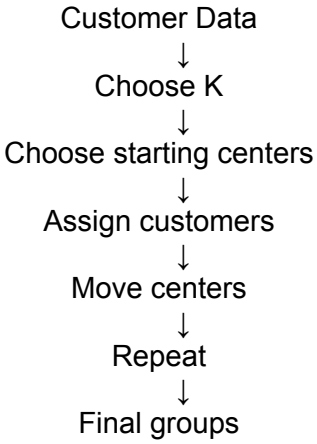
Step 1: Choose K, you decide how many clusters you want the algorithm to create. The algorithm needs this number so it knows how many groups to look for. For example, if you choose K = 3, you are telling the algorithm to find three groups in the data.

Step 2: Initialize centers, the algorithm picks starting positions for the cluster centers. These starting points are often chosen randomly. Because the starting positions can be different each time, the final clusters can also end up slightly different, especially when the data has a lot of variation.

Step 3: Assign data points, each data point is matched to the closest center. The algorithm measures the distance from the point to each center and assigns the point to the one that is nearest. Points that are closest to the same center become part of the same cluster.

Step 4: Update and repeat, the algorithm recalculates each center by averaging the points in that cluster. Then it repeats the assignment step using the new centers. This cycle continues until the centers stop changing much or the algorithm reaches its limit.

Flowchart Visual:



5. Applying K-Means to Customer Segmentation

In a real business, K-Means helps a company understand customers by looking at simple numbers. These numbers might be how much someone spends, how many things they buy, or how often they shop. The algorithm compares these numbers and looks for customers who seem alike. When people have similar shopping patterns, K-Means puts them in the same group.

The algorithm doesn't know what the groups mean. It doesn't automatically say, "This group is high spenders" or "This group shops a lot." A person has to look at the groups afterward and decide what each one represents. The meaning depends on the business and the type of customer data they collected.

Once the clusters are understood, they become very useful. A business can use these clusters to understand different types of customers. For example, they might find a group of loyal shoppers or a group that only buys during sales. Knowing this helps the business make better choices, like creating targeted marketing, offering the right promotions, or focusing on valuable customer segments. *Madhu et al. (2021)* explain that K-Means is useful because it reveals patterns that are hard to see just by looking at raw numbers.

6. Choosing K and Evaluating Cluster Quality

In K-Means, we need to choose how many clusters we want before the algorithm starts grouping the data. For example, if we choose K = 3, the data will be divided into three groups. Choosing the right number of clusters matters because too few mix different things together, and too many split things that belong together.

The **elbow method** helps us figure out how many clusters to use. We look at a graph that shows the number of clusters and the inertia. We look for the point where the line starts to bend, like an elbow. After that point, adding more clusters does not make as much of a difference.

Inertia, also called SSE, tells us how close the data points are to the center of their groups. Lower inertia means the points are closer to their group centers. However,

adding more clusters usually lowers inertia, so the lowest number does not always mean we have the best result.

The **silhouette score** helps us see how well the groups are separated from each other. A higher score usually means the groups are better separated. Looking at both inertia and the silhouette score helps us make a better decision instead of using just one. We could look at the actual graphs before deciding which number of clusters works best.

7. What Recent Research Says About K-Means

Liu and Huang (2025) studied different versions of the K-Means algorithm in their paper *A Survey on K-Means Clustering Algorithms: Theoretical Analysis and Performance Comparison*. They compared how well these versions form clusters and how fast they run. Their main point is that each version has strengths and weaknesses. Some create better clusters, while others finish more quickly. Because of this, choosing the right version depends on the type of data and what the user wants to accomplish.

Madhu et al. (2021), in *Customer Segmentation Using K-Means Clustering*, used K-Means on online shopping data from a UK store. They compared K-Means with hierarchical clustering to see how each method groups customers. Their study shows that clustering can help businesses understand customer behavior, such as spending patterns or shopping frequency. However, the results depend on having good, clean customer data and choosing useful features to analyze.

Ikotun et al. (2023), in *K-means Clustering Algorithms: A Comprehensive Review, Variants Analysis, and Advances in the Era of Big Data,* reviewed many versions of K-Means and ways to improve them. They explained common challenges, such as picking the right number of clusters and choosing good starting points. These choices can change the final results, which shows that K-Means is helpful but not perfect.

These three papers show that K-Means is useful, but the method must be chosen carefully. The 2025 paper compares different versions, the 2021 paper shows how clustering helps businesses understand customers, and the 2023 paper explains the challenges researchers are trying to solve.

In my opinion, K-Means can help businesses group customers based on their shopping habits. But I would not trust the results without checking the clusters first. Businesses should make sure their data is accurate and choose the number of clusters wisely so they can make better decisions about marketing and customer needs.

8. Limitations & Alternative Methods

K-Means is useful for grouping data, but it has some limits. One challenge is choosing the number of clusters, called K. If we choose the wrong number, the groups may not be helpful.

K-Means can also give different results depending on where the starting centers are placed. It can be affected by outliers, which are data points that are very different from the others. These points can make the groups less accurate.

K-Means works best with numbers, such as customer income and spending scores. It does not work directly with categories, such as product types or favorite colors.

Other methods can be better for certain types of data. K-Medoids uses actual data points as group centers, which can help reduce the effect of outliers. K-Modes work with categories instead of numbers. Hierarchical clustering builds groups step by step and shows how they are related.

After reading the 2025 survey by Liu and Huang, I learned that different versions of K-Means have been created to address some of its problems. Overall, I believe the best method depends on the data and what we want to learn. K-Means is a good starting point, but it is not always the best choice.

9. Experiment & What I learned
Focus: Annual Income (k$) and spending score (1-100)

A. Dataset and Objective

For my experiment, I used the Mall Customer Segmentation Dataset. It contains 200 customer records, and I selected annual income and spending scores to study. My goal was to see how K-Means could group similar customers based on their shopping habits and spending. I also wanted to understand how changing the number of clusters affects the results.

B. Method

Before running K-Means, I checked the selected features for missing or invalid values. All 200 records were retained. I used feature scaling to put both features on a similar scale and tested K = 2, 3, 4, and 5. To compare the results, I used the elbow method, inertia, and silhouette score. I also created graphs to visualize the customer groups.

C. Results

My elbow graph showed a large decrease in inertia between K = 2 and K = 3, after which the curve began to flatten, although the elbow was not completely clear. The highest silhouette score was 0.5547 at K = 5. My cluster visualizations showed groups with high income and low spending, low income and high spending, and low income and low spending. The elbow method and silhouette score did not agree on the preferred number of clusters.

D. Interpretation

The clusters showed different customer spending patterns. For example, Cluster 3 had a mean annual income of approximately $88,200 but a low spending score of 17.11, while Cluster 2 had a mean annual income of approximately $25,727 and a high spending score of 79.36. These results suggest that income alone does not explain customer spending behavior. Businesses could use these patterns to better understand their customers, but they should examine the data further before making marketing decisions.

E. Personal Takeaways
This experiment helped me understand that choosing the number of clusters is not always straightforward. I learned that different evaluation methods can suggest different results and that more clusters do not automatically mean better business insights. If I

repeated the experiment, I would test additional K values, include more customer features, and compare K-Means with other clustering methods.

10. Conclusion

K-Means is a machine learning method that helps businesses group customers based on their spending and shopping habits. These groups help businesses understand their customers, find shopping patterns, and create better marketing strategies.

My research showed that K-Means has different versions, each with its own strengths. It also showed how grouping customers can help businesses learn more about shopping behavior.

Choosing the right number of groups, preparing the data, and checking the results are important for getting useful results. Overall, K-Means is a helpful tool for businesses, but they should review the results before making decisions about customers and marketing.

References

Ikotun, A. M., Ezugwu, A. E., Abualigah, L., Abuhaija, B., & Heming, J. (2023). K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data. *Information Sciences, 622*, 178–210. https://www.sciencedirect.com/science/article/abs/pii/S0020025522014633

Kavlakoglu, E., & Winland, V. (2024, June 26). *What is K-means clustering?* IBM. https://www.ibm.com/think/topics/k-means-clustering

Liu, Y., & Huang, S. (2025). *A survey on K-means clustering algorithms: Theoretical analysis and performance comparison*. arXiv. https://knightnemo.github.io/assets/files/clustering.pdf

Madhu, J., Revanakar, K., Lavanya, & Akash. (2021). Customer segmentation using K-means clustering. *International Journal of Advances in Engineering and Management, 3*. https://ijaem.net/issue_dcp/Customer%20Segmentation%20using%20K%20means%20Clustering.pdf

Silva, M. De. (2024, November 18). *A beginner's guide to clustering: Customer segmentation with K-means clustering*. Medium. https://medium.com/@maleeshadesilva21/a-beginners-guide-to-clustering-customer-segmentation-with-k-means-clustering-c4e35c527ef8

Seenivasa Ramadurai. (2025, February 6). *Segmenting customers with K-means clustering: A data-driven approach to targeted marketing*. DEV Community. https://dev.to/sreeni5018/title-segmenting-customers-with-k-means-clustering-a-data-driven-approach-to-targeted-marketing-328n